A Confirmed Pattern Without a Confirmed Cause: Group Delay Dispersion Separates Synthetic Speech More in Sonorants, and So Does a Resampler

Darwin Zhang

Abstract

We pre-registered and confirmed that the dispersion of Modified Group Delay over the upper spectrum separates genuine from synthetic speech more strongly in sonorants than in obstruents, relative to a cepstral statistic, on ASVspoof 2019 LA: a difference of $+0.063$ [+0.045, +0.080] in folded AUC. The natural reading is a phase artifact of vocoding, concentrated where speech is periodic. We then tried to break that reading, and it does not hold. A low quality sample rate conversion applied to genuine speech, with no synthesis involved, reproduces the pattern at full size. A magnitude statistic, spectral tilt, tracks it. The pattern itself is robust: it survives resampling of utterances, speakers and synthesis systems, and an identical low pass on both classes, on two partitions. We report it as a confirmed observation whose cause is undecided, and describe the checks that changed our own conclusion.

1 Introduction

Detectors for synthetic speech are strong (Jung et al., 2022), and say little about where in speech their evidence lies. A complementary question is descriptive: is the difference between genuine and synthetic speech spread evenly across classes of sound? If a vocoder must reconstruct harmonic phase, one would expect an artifact where speech is voiced and periodic, and little where it is noise-like.

We test the pattern that idea predicts with one fixed pair of statistics, under pre-registration, and then subject the result to the checks a sceptic would ask for. This paper is mostly about those checks. Separations here run from 0.51 to 0.60 AUC. Nothing in it is a detector.

2 Method

Data. ASVspoof 2019 Logical Access (Wang et al., 2020). The evaluation partition supplies 1,500 bonafide utterances from 67 speakers and 115 utterances from each of 13 synthesis systems, A07 to A19. The development partition, with systems A01 to A06, is used only for a second registered test.

Segmentation. A multilingual phone recogniser (Xu et al., 2022) gives a frame-level phone sequence. Each run of a phone becomes a slice, assigned to one of six categories: nasals, approximants and steady vowels (sonorants), and fricatives, plosives and affricates (obstruents). This yields 68,996 slices on the evaluation partition. The segmentation is approximate: 89% of slices are a single 20 ms frame widened to a 60 ms floor.

Statistics. The phase statistic, MGD-HF, is the standard deviation of Modified Group Delay (Murthy and Gadde, 2003) over the upper half of the spectrum, 4 to 8 kHz. The magnitude statistic is the spread of MFCC frames within the slice. Both were fixed before the evaluation data was examined.

Quantities. For a statistic and a category, separation is the folded AUC, $|\mathrm{AUC} - 0.5| + 0.5$. The *advantage* in a category is the separation of MGD-HF minus that of the magnitude statistic. The *level* is the mean advantage over the three sonorant categories. The *asymmetry* is the level minus the mean advantage over the three obstruent categories. The registered hypothesis, H1, is that the asymmetry is positive with a 95% interval excluding zero.

Inference. Slices are not independent, so intervals come from a cluster bootstrap (Efron and Tibshirani, 1993) with 2,000 draws in which one resample of clusters serves every category in a draw. The registered cluster is the source utterance. A percentile bound is itself an estimate, so each bound carries a distribution-free Monte Carlo band, and a verdict that turns on a bound inside that band is reported as undecided.

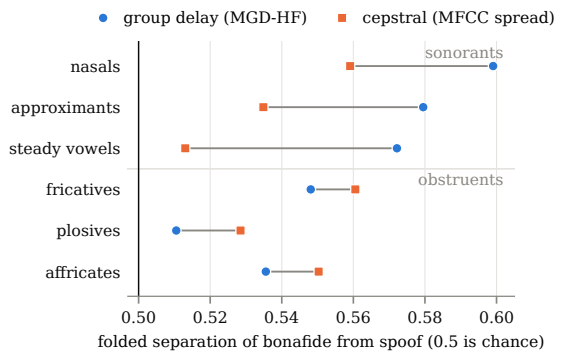


Figure 1: Separation of bonafide from spoofed speech by category, evaluation partition. The group delay statistic leads in sonorants and trails in obstruents.

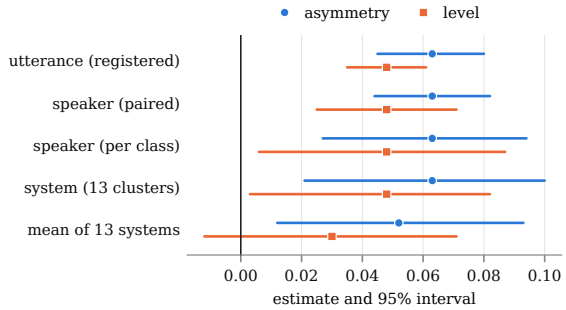


Figure 2: The same two quantities under different choices of what the bootstrap resamples, and as a mean over the 13 systems scored one at a time.

3 The confirmed pattern

Figure 1 shows the pattern. The level is $+0.048$ [+0.035, +0.061] and the asymmetry $+0.063$ [+0.045, +0.080]. All registered falsification conditions were decided in favour of H1.

4 How far it reaches

What is resampled. Clustering by utterance describes more utterances from the same speakers and systems. Figure 2 asks more. The asymmetry excludes zero whether speakers or synthesis systems are resampled. The level does not: with 13 system clusters its lower bound sits on zero, and scored one system at a time its mean is $+0.030$ [-0.012, +0.071]. The pooled level is inflated because the magnitude statistic runs in different directions on different systems, which cancels in the pool.

Which statistics. Figure 3 swaps one side of the pair at a time. These are sensitivity analyses on data already seen, not tests. Two things follow. The asymmetry belongs to upper-band group

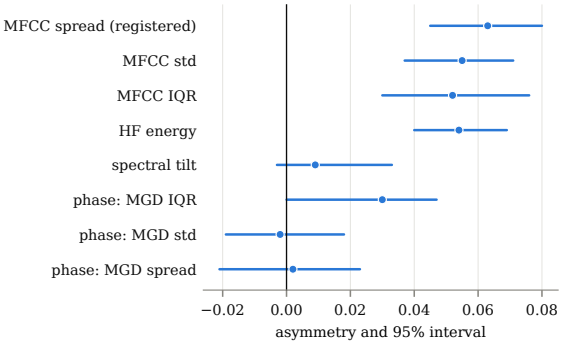


Figure 3: The asymmetry under other pairs of statistics. The first five rows keep MGD-HF and change the magnitude comparator. The last three keep the registered comparator and change the phase statistic to a whole-band one. It holds against cepstral and energy statistics and vanishes against spectral tilt.

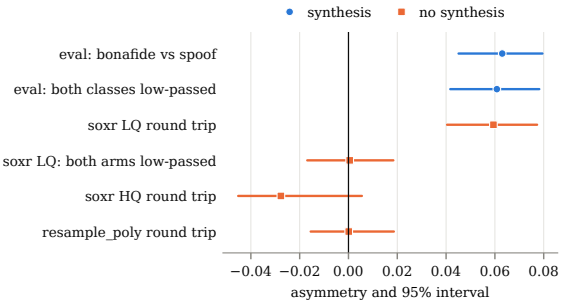


Figure 4: Asymmetry for the real comparison and for genuine speech against the same speech after a 16 to 22.05 to 16 kHz round trip. One resampler reproduces the effect with no synthesis. An identical low pass removes that artifact and leaves the real one.

delay dispersion, not to phase statistics in general. And with spectral tilt as the comparator it is $+0.009$ [-0.003, +0.033]: a magnitude statistic with no phase content shows nearly the same sonorant-over-obstruent pattern.

5 Is it synthesis at all?

An anti-alias filter is a group delay phenomenon at the band edge, which is what MGD-HF reads. We wrote three decision rules before reading any audio, then compared each bonafide utterance with itself after a sample rate round trip, rounded back to 16 bit (Figure 4).

With a low quality resampler, whose passband rolls off from about 6 kHz, the asymmetry is $+0.059$ [+0.040, +0.077] and the level $+0.055$: the size, direction and category order of the real result. Two sharper resamplers give nothing. So the signature of H1 is not specific to synthesis, and

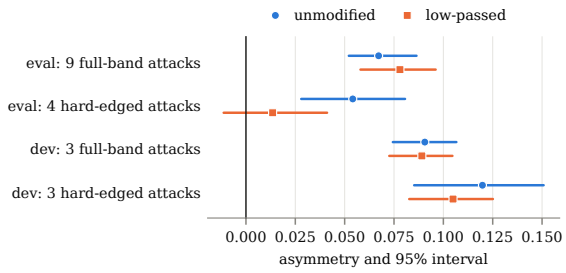


Figure 5: A split seen after the fact on the evaluation partition, and its registered test on the development partition, where it does not replicate.

the rule written to detect that fires.

The same artifact is nonetheless not what the corpus shows. One identical linear-phase low pass at 6.9 kHz on both arms takes the planted asymmetry to $+0.001$ [-0.017, +0.018], and leaves the real one at $+0.061$ [+0.042, +0.078]. The low pass cannot rule out a difference in spectral shape below its cutoff, and the tilt result suggests there is one.

A post hoc split, tested. Four evaluation systems carry a hard band edge near 7.7 kHz, and pooled over them the asymmetry did not survive the low pass (Figure 5). Because that split was chosen after seeing the data, we registered it and tested it on the development partition before reading its audio. Three of its six systems are hard-edged. Pooled over those three the asymmetry is $+0.105$ [+0.083, +0.125] after the low pass, so the split does not replicate. An independent re-derivation reproduced these numbers and narrowed their reading: the pool is positive only because of one system (A01), the other two have a negative asymmetry that the low pass moves toward zero, and over all six systems the low pass lowers the asymmetry by 0.033 [0.027, 0.039]. Band edges matter on this partition; the pooled prediction about them failed.

6 What we got wrong

Four errors were caught by re-derivation rather than by review of our own reasoning, and each changed a published number or sentence. (i) Intervals on the level were first computed by resampling each category independently, which ignores that one utterance feeds all three; they were about 40% too narrow. (ii) Under system clustering the printed verdict changed with the random seed. (iii) The registered primary test, the interval on the asymmetry, was printed but not used in the verdict. (iv) We first set aside the one resampler that reproduces the effect, on the ground that it also moves the magnitude statistic, which the real comparison does too.

7 Limitations

One corpus and one language. Thirteen systems and 67 speakers. Percentile intervals, which cover less than their nominal rate with 13 clusters. A segmentation that is close to fixed windows. The development result was re-derived independently once; the size of the change under the low pass depends on the filter cutoff and on whether affricates are counted. The statistic pair was chosen on an earlier partition as the strongest of several, and the result is specific to it.

8 Conclusion

The pattern is real in the sense that matters for a registered claim: it was predicted, it appeared, and it survives changes of resampling unit, an equalising low pass and a second partition. Its interpretation as a phase artifact of synthesis is not supported. A resampler produces the same signature, and spectral tilt carries the same pattern. A control in which magnitude is real and phase is wholly synthetic is registered and not yet run; it is the test that can separate the two accounts.

Reproducibility

Code, the slice tables, registrations and the commands behind every number are in the project repository. Figures are drawn by `scripts/make_report_figures.py`. The detailed records are `docs/ROBUSTNESS.md`, `docs/BAND-EDGE.md` and the registration documents.

References

Bradley Efron and Robert J. Tibshirani. 1993. *An Introduction to the Bootstrap*. Chapman & Hall.

Jee-weon Jung, Hee-Soo Heo, Hemlata Tak, Hye-jin Shim, Joon Son Chung, Bong-Jin Lee, Ha-Jin Yu, and Nicholas Evans. 2022. AASIST: Audio anti-spoofing using integrated spectro-temporal graph attention networks. In *Proceedings of ICASSP*.

Hema A. Murthy and Venkata Gadde. 2003. The modified group delay function and its application to phoneme recognition. In *Proceedings of ICASSP*.

Xin Wang, Junichi Yamagishi, Massimiliano Todisco, Héctor Delgado, Andreas Nautsch, Nicholas Evans, et al. 2020. ASVspoof 2019: A large-scale public database of synthesized, converted and replayed speech. *Computer Speech & Language*, 64:101114.

Qiantong Xu, Alexei Baevski, and Michael Auli. 2022. Simple and effective zero-shot cross-lingual phoneme recognition. In *Proceedings of Interspeech*.